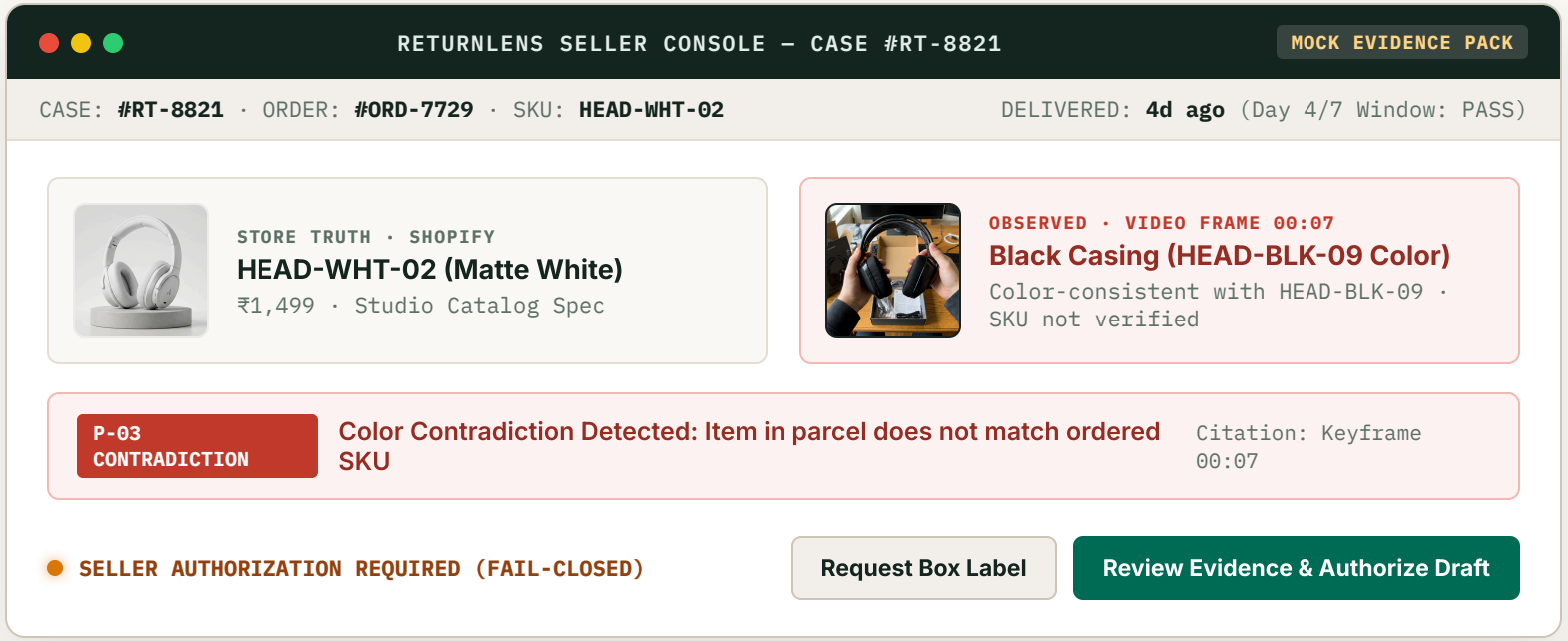
ReturnLens

Every return decision, backed by verifiable evidence.

ReturnLens unifies unboxing video evidence, store truth, and policy checks into a 1-click decision console for solo merchants.

> **Core Thesis:** Models observe. Code verifies. Seller authorizes.
> Scope: Return triage + order status · Mock fixtures · Never auto-refunds



THE PROBLEM SOLO MERCHANT BOTTLENECK

One return. Four disconnected sources.

Solo merchants cross-reference tabs, video footage, and policy rules manually before moving money.

SOURCE 01
Buyer Message
Subjective text & unindexed video claim in chat

SOURCE 02
Order Record
Shopify manifest: HEAD-WHT-02, Matte White, ₹1,499

SOURCE 03
Courier Status
Shippo scan: delivered 4 days ago (Day 4 of 7)

SOURCE 04
Return Policy
7-day defect window; security packaging tag rule

THE MISSING LINK

“What evidence supports this return decision?”

The gap is the absence of an auditable artifact linking order truth, video keyframes, and policy before money moves.

9% vs 15.14%

Industry Baseline: 9.0% Return Fraud (NRF 2025, criminal fraud only) vs 15.14% Fraud + Policy Abuse (Appriss/Deloitte 2024) (Definitions differ).

Manual Triage

Asymmetric Merchant Risk: High cognitive overhead on unboxing triage; memory-based manual checks create friction and financial leaks.

STAR DEMO · CASE B · CORE BENCHMARK

The order says white. The video shows black.

ReturnLens flags the contradiction — the seller decides.

STORE CATALOG TRUTH — #ORD-7729

SKU: **HEAD-WHT-02**
Color: **Matte White**
Price: **₹1,499 paid**
Delivered: **4d ago (Shippo #SH-9012)**

≠

COLOR FAIL

MOCK EVIDENCE PACK · VIDEO CITATION — FRAME 00:07



Observed: **Black Casing**
Color Match: **Consistent with HEAD-BLK-09**
Exact SKU: **Not verified (Barcode unverified)**
Lighting: **Optimal (142/255) · REC 00:07 / 00:24**

POLICY P-03 RESULT

ESCALATE TO SELLER

3 Preserved Seller Actions: (1) Accept as warehouse mis-shipment · (2) Request photo of outside-box shipping label · (3) Hold return for physical parcel inspection.

Server Cap Formula:
`0 < amount ≤ min(₹4,999, rem, paid)`

SYSTEM ARCHITECTURE FAIL-CLOSED GOVERNANCE

Models observe. Code verifies. Seller authorizes.

Dynamic action selection and stateful resumption under deterministic human authorization.

1 **Planner** Typed actions → 2 **Tools** FFmpeg ~1fps + APIs → 3 **Policy P-01-08** Deterministic rules → 4 **Critic** Review cited frames → 5 **Seller Gate** 1-Click authorization → 6 **Audit Log** Append-only JSON

CALL BUDGET

Exactly 3 Model Calls

1. Planner proposes typed tool actions.
2. Vision model inspects server-generated keyframes.
3. Critic evaluates citations against policy. Model never accesses payment APIs directly.

STATEFUL RESUMPTION

Case C: Re-Film Loop

When video is too dark (brightness 28, 2 frames), agent calls `ask_for_evidence()` with a re-film checklist. Case state saved to `WAITING_FOR_EVIDENCE` and resumes on upload. After 1 retry, escalates to seller with both attempts logged (never auto-denies).

SECONDARY ROUTE

Tiny PS-04 Order Route

For routine *"Where is my order?"* inquiries, agent bypasses vision entirely: Store API → Courier API → Instant status reply. Zero token waste.

FAIL-CLOSED FINANCIAL GATE: `requires_seller_approval: true` · Approval binds case + evidence hash + policy version. Server Cap: `0 < amount <= min(₹4,999, remaining_refundable, order_paid)` · Revalidation check before draft release.

EVALUATION SUITE 4 CONTROLLED SCENARIOS

Different evidence. Different action.

Every case produces an auditable mock evidence pack. No automatic policy approvals.

CORE BENCHMARK SCENARIO

ESCALATE

Case B: Color Contradiction

P-03 Failure • Order White vs Video Black

Order Record:	**HEAD-WHT-02 (Matte White, ₹1,499)**
Observed Frame 00:07:	**Black casing (color-consistent with HEAD-BLK-09, SKU not verified)**
Policy Evaluation:	**P-03 Color Match: FAIL**
Automated Action:	No label • No refund draft • No denial

PRESERVED SELLER OPTIONS IN MOCK EVIDENCE PACK:

- Accept as warehouse mis-shipment
- Request customer photo of outer box shipping barcode
- Hold case for physical receipt inspection at depot

Case A: Valid Return (ORD-1042)

PREPARE SELLER REVIEW

Evidence: ORD-1042 • ₹1,499 • SKU matches catalog • Tag clearly present • Within 7-day return window.

Outcome: Mock shipping label created • Refund draft placed in pending_return_receipt (not executed).

Case C: Low-Light Video (Re-Film Loop)

ASK & RESUME

Evidence: 4-second dark unboxing video • Brightness score 28/255 • Only 2 usable keyframes extracted.

Outcome: Agent issues re-film checklist • Case stays in WAITING_FOR_EVIDENCE • Resumes on upload (after 1 retry, escalates with both attempts logged; never auto-denies).

Case D: Continuous Unboxing Sequence

INSPECT SEQUENCE

Evidence: Single continuous take from opening sealed parcel to displaying label in one sequence.

Outcome: Inspects sequence continuity and flags abrupt transitions. *Does not claim proof of identical item.*

24-HOUR PROTOTYPE PLAN TEAM ROOKIE (3 MEMBERS)

A credible prototype in 24 hours.

Strict scope boundaries and single-threaded ownership make the prototype buildable in one day.

00 – 06h
Foundation
LangGraph state machine, mock store & courier fixtures, order status fast path.

06 – 12h
Evidence
FFmpeg keyframe indexer, vision prompt schema, critic prompt, seller console UI.

12 – 18h
Integration
Case C pause/resume loop · 3 core cases (A–C) + Case D sequence test · Critic review

18 – 24h
Validation
Authorization safety tests, edge cases, fallback validation, demo video recording.

PERSON A · BACKEND, POLICY & AUDIT

State Machine & Policy Engine

`Stack: LangGraph · Pydantic · SQLite · SHA256`

- Policy P-01 to P-08: Deterministic rules (window, SKU, tag, cap)
- Critic Agent: Adversarial check against cited timestamps
- Audit Ledger: Append-only JSON log & approval binding
- Financial Cap: Server calculation min(₹4,999, rem, paid)

PERSON B · VISION & CONSOLE

Video Pipeline & Seller UI

`Stack: FFmpeg · Hosted vision model (unpinned) · HTML/CSS`

- Keyframe Pipeline: Uniform ~1fps sampling & brightness (28/255)
- Structured Vision: Observation schema with timestamp citations
- Seller Console: 1-click decision console with mock pack
- Grounding Guard: Enforces unverified barcode vs color check

PERSON C · INTEGRATION & QA

Scenarios, Lifecycle & QA

`Stack: Pytest · Staged Fixtures · Automated Smoke QA`

- Benchmark Cases: Fixtures for Cases A–C + Case D sequence
- Case C Lifecycle: Re-film loop in WAITING_FOR_EVIDENCE (1 retry)
- Fail-Closed Testing: Verifies zero automated refunds & gates
- Automated QA: 9-test smoke runner & video demo recording

MUST DELIVER (CORE PROTOTYPE)

Mock store & courier APIs • Deterministic policy engine & seller gate • Case B color contradiction & Case C re-film loop • 3 core demo cases (A–C) + Case D sequence test • Approval binding (case + evidence hash + policy version) • Structured JSON audit trail & order-status fast path

CUT FIRST IF BEHIND

Advanced video forensic splice analysis (Case D stretch) • WhatsApp bot • Exchanges • Multi-seller auth • Audio analysis • Public cloud deployment

FINANCIAL CEILING FORMULA

$0 < amt \leq min(₹4{,}999, rem_ref, paid)$